In [1]:
```
import pandas as pd

dnkn = pd.read_csv("BUS_303_Dunkin_Donuts_Nutritional_Menu_Kaggle_2023.csv")

dnkn
```

Out[1]:

	Category	Item	Calories	Total Fat (g)	Saturated Fat (g)	Trans Fat (g)	Cholesterol (mg)	Sodium (mg)	Carbs (g)	Fiber (g)	Sugars (g)	Protein (g)	Weight Watcher Pnts
0	Cold Brew Coffee	Cold Brew Coffee, Large	5	0	0	0	0	20	0	0	0	1	4
1	Cold Brew Coffee	Cold Brew Coffee, Medium	5	0	0	0	0	15	0	0	0	0	5
2	Cold Brew Coffee	Cold Brew Coffee, Small	5	0	0	0	0	10	0	0	0	0	5
3	Cold Brew Coffee	Cold Brew Coffee with Cream, Large	120	12	6	0	35	65	2	0	2	2	126
4	Cold Brew Coffee	Cold Brew Coffee with Cream, Medium	90	9	4.5	0	25	45	1	0	1	2	93.5
...	...	...	...	...	...	...	...	...	...	...	...	...	...
775	Soft Serve	OREO Mix-In	510	21	11	0.5	55	350	70	2	55	14	562
776	Soft Serve	Soft Serve Cone	260	9	6	0	35	160	35	1	27	9	284
777	Soft Serve	Strawberry Coolatta Float	420	12	8	0	50	210	68	1	64	11	481
778	Hash Browns	Hash Browns	130	6	1	0	0	420	12	2	0	5	126

	Category	Item	Calories	Total Fat (g)	Saturated Fat (g)	Trans Fat (g)	Cholesterol (mg)	Sodium (mg)	Carbs (g)	Fiber (g)	Sugars (g)	Protein (g)	Weight Watcher Pnts
779	Dunkin' Bowls	Egg White Bowl	250	11	4	0	15	490	20	0	3	14	243

780 rows × 13 columns

In [2]:
```
dnkn.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 780 entries, 0 to 779
Data columns (total 13 columns):
 #   Column               Non-Null Count  Dtype 
---  ------               --------------  ----- 
 0   Category             780 non-null    object
 1   Item                 780 non-null    object
 2   Calories             780 non-null    object
 3   Total Fat (g)        780 non-null    object
 4   Saturated Fat (g)    780 non-null    object
 5   Trans Fat (g)        780 non-null    object
 6   Cholesterol (mg)     780 non-null    object
 7   Sodium (mg)          780 non-null    object
 8   Carbs (g)            780 non-null    object
 9   Fiber (g)            780 non-null    object
 10  Sugars (g)           780 non-null    object
 11  Protein (g)          780 non-null    object
 12  Weight Watcher Pnts  780 non-null    object
dtypes: object(13)
memory usage: 79.3+ KB
```

In [3]:
```
dnkn.isnull().sum()
```

Out[3]:
```
Category              0
Item                  0
Calories              0
Total Fat (g)         0
Saturated Fat (g)     0
Trans Fat (g)         0
Cholesterol (mg)      0
Sodium (mg)           0
Carbs (g)             0
Fiber (g)             0
Sugars (g)            0
Protein (g)           0
Weight Watcher Pnts   0
dtype: int64
```

In [4]:
```
dnkn.duplicated().sum()
```

Out[4]:
```
1
```

In [5]:
```
dnkn[dnkn.duplicated(keep=False)]
```

Out[5]:

	Category	Item	Calories	Total Fat (g)	Saturated Fat (g)	Trans Fat (g)	Cholesterol (mg)	Sodium (mg)	Carbs (g)	Fiber (g)	Sugars (g)	Protein (g)	Weight Watcher Pnts
737	Sandwiches	Ham and Cheese Rollups	280	13	6	0	40	1050	27	1	2	13	275
760	Sandwiches	Ham and Cheese Rollups	280	13	6	0	40	1050	27	1	2	13	275

In [6]:
```
dnkn.nunique()
```

Out[6]:
```
Category               23
Item                  779
Calories               87
Total Fat (g)          50
Saturated Fat (g)      29
Trans Fat (g)           5
Cholesterol (mg)       37
Sodium (mg)           112
Carbs (g)             120
Fiber (g)               9
Sugars (g)            122
Protein (g)            32
Weight Watcher Pnts   362
dtype: int64
```

In [7]:
```
dnkn['Category'].value_counts()
```

Out[7]:
```
Category
Hot Coffee            88
Donuts                87
Hot Latte             63
Iced Coffee           63
Iced Latte            63
Sandwiches            55
Frozen Coffee         49
Hot Macchiato         45
Hot Cappuccino        43
Iced Macchiato        42
Iced Cappuccino       42
Iced Tea              33
Hot Chocolate         28
Muffins               18
Kolache               16
Cold Brew Coffee      11
Coolatta               9
Dunkin’ Refreshers     9
Soft Serve             8
Hot Americano          3
Iced Americano         3
Hash Browns            1
Dunkin’ Bowls          1
Name: count, dtype: int64
```

In [8]:
```
dnkn.describe(include='all')
```

Out[8]:

	Category	Item	Calories	Total Fat (g)	Saturated Fat (g)	Trans Fat (g)	Cholesterol (mg)	Sodium (mg)	Carbs (g)	Fiber (g)	Sugars (g)	Protein (g)	Weight Watcher Pnts
count	780	780	780	780	780	780	780	780	780	780	780	780	780
unique	23	779	87	50	29	5	37	112	120	9	122	32	362
top	Hot Coffee	Ham and Cheese Rollups	170	0	0	0	0	180	35	0	0	4	5
freq	88	2	38	242	269	755	222	24	30	508	33	94	13

In [9]:

```
columns_to_check = [
    'Total Fat (g)',
    'Saturated Fat (g)',
    'Trans Fat (g)',
    'Cholesterol (mg)',
    'Sodium (mg)',
    'Fiber (g)',
    'Protein (g)',
    'Carbs (g)',
    'Sugars (g)'
]

for column in columns_to_check:
    print(f"\n{column}:")
    print(dnkn[column].unique())
```

```
Total Fat (g):
['0' '12' '9' '6' '4.5' '3' '10' '11' '28' '30' '22' '21' '23' '14' '15'
 '20' '19' '16' '18' '25' '31' '27' '17' '37' '40' '5' '8' '35' '26' '13'
 '7' '\xa0' '4' '1' '0.5' '1.5' '2.5' '2' '3.5' '42' '33' '24' '36' '54'
 '29' '32' '34' '52' '41' '43']

Saturated Fat (g):
['0' '6' '4.5' '3' '3.5' '2.5' '1.5' '4' '12' '14' '9' '10' '11' '7' '5'
 '8' '13' '18' '22' '17' '15' '\xa0' '0.5' '19' '21' '1' '2' '20' '16']

Trans Fat (g):
['0' '1.5' '1' '0.5' '\xa0']

Cholesterol (mg):
['0' '35' '25' '20' '15' '10' '50' '30' '40' '130' '100' '70' '105' '80'
 '55' '<5' '\xa0' '125' '45' '60' '185' '340' '180' '95' '65' '190' '360'
 '195' '265' '175' '220' '120' '200' '240' '205' '110' '115']

Sodium (mg):
['20' '15' '10' '65' '45' '30' '5' '60' '320' '240' '160' '300' '310'
 '470' '450' '420' '150' '340' '290' '400' '390' '440' '280' '135' '380'
 '480' '270' '430' '460' '360' '200' '370' '190' '330' '250' '170' '220'
 '180' '130' '230' '125' '100' '115' '75' '350' '260' '120' '\xa0' '95'
 '70' '50' '25' '85' '55' '90' '140' '105' '110' '210' '510' '570' '630'
 '500' '750' '620' '80' '40' '35' '1020' '810' '1260' '1460' '600' '410'
 '840' '1060' '800' '820' '1170' '520' '540' '670' '640' '1200' '590'
 '1190' '1210' '1080' '1240' '1980' '1010' '650' '960' '1140' '1220'
 '1270' '900' '910' '1050' '980' '1500' '1120' '710' '550' '850' '490'
 '760' '970' '580' '1160' '700' '680']

Fiber (g):
['0' '1' '2' '3' '4' '\xa0' '6' '5' '8']

Protein (g):
['1' '0' '2' '10' '7' '5' '4' '6' '3' '8' '11' '9' '\xa0' '12' '16' '15'
 '14' '13' '27' '24' '37' '17' '19' '26' '21' '23' '18' '28' '33' '22'
 '25' '20']

Carbs (g):
['0' '2' '1' '37' '27' '18' '112' '84' '56' '172' '129' '86' '113' '85'
 '31' '44' '57' '47' '63' '39' '55' '48' '36' '58' '33' '41' '34' '50'
 '40' '21' '42' '61' '43' '32' '52' '38' '59' '35' '45' '30' '49' '24'
```

```
 '176' '135' '95' '175' '134' '94' '180' '139' '97' '184' '142' '99' '141'
 '138' '183' '117' '88' '177' '160' '120' '80' '\xa0' '191' '147' '103'
 '195' '150' '105' '111' '83' '110' '13' '10' '7' '65' '64' '73' '70' '53'
 '71' '51' '87' '72' '66' '89' '82' '12' '9' '6' '26' '67' '5' '4' '46'
 '3' '28' '19' '17' '25' '23' '16' '15' '79' '78' '60' '11' '20' '54' '14'
 '69' '22' '75' '77' '29' '74' '68' '8' '92']

Sugars (g):
['0' '2' '1' '36' '27' '18' '110' '83' '55' '167' '125' '107' '80' '53'
 '10' '22' '24' '21' '11' '34' '32' '26' '14' '33' '20' '13' '15' '19'
 '16' '28' '17' '23' '12' '25' '5' '29' '165' '126' '88' '162' '124' '87'
 '158' '121' '84' '172' '132' '92' '176' '135' '94' '175' '134' '169'
 '130' '91' '173' '133' '93' '108' '81' '54' '112' '56' '111' '159' '119'
 '151' '113' '75' '\xa0' '181' '139' '97' '185' '142' '99' '98' '109' '82'
 '9' '6' '47' '35' '46' '63' '31' '61' '30' '60' '45' '58' '43' '57' '38'
 '62' '50' '39' '70' '42' '71' '65' '69' '64' '51' '66' '40' '49' '37' '7'
 '3' '4' '59' '48' '74' '52' '68' '67' '78' '8' '44' '85']
```

In [10]:
```
dnkn1 = dnkn.copy()
dnkn1
```

Out[10]:

	Category	Item	Calories	Total Fat (g)	Saturated Fat (g)	Trans Fat (g)	Cholesterol (mg)	Sodium (mg)	Carbs (g)	Fiber (g)	Sugars (g)	Protein (g)	Weight Watcher Pnts
0	Cold Brew Coffee	Cold Brew Coffee, Large	5	0	0	0	0	20	0	0	0	1	4
1	Cold Brew Coffee	Cold Brew Coffee, Medium	5	0	0	0	0	15	0	0	0	0	5
2	Cold Brew Coffee	Cold Brew Coffee, Small	5	0	0	0	0	10	0	0	0	0	5
3	Cold Brew Coffee	Cold Brew Coffee with Cream, Large	120	12	6	0	35	65	2	0	2	2	126
4	Cold Brew Coffee	Cold Brew Coffee with Cream, Medium	90	9	4.5	0	25	45	1	0	1	2	93.5
...	...	...	...	...	...	...	...	...	...	...	...	...	...
775	Soft Serve	OREO Mix-In	510	21	11	0.5	55	350	70	2	55	14	562
776	Soft Serve	Soft Serve Cone	260	9	6	0	35	160	35	1	27	9	284
777	Soft Serve	Strawberry Coolatta Float	420	12	8	0	50	210	68	1	64	11	481
778	Hash Browns	Hash Browns	130	6	1	0	0	420	12	2	0	5	126

	Category	Item	Calories	Total Fat (g)	Saturated Fat (g)	Trans Fat (g)	Cholesterol (mg)	Sodium (mg)	Carbs (g)	Fiber (g)	Sugars (g)	Protein (g)	Weight Watcher Pnts
779	Dunkin' Bowls	Egg White Bowl	250	11	4	0	15	490	20	0	3	14	243

780 rows × 13 columns

In [11]:
```
food_categories = [
    'Donuts', 'Sandwiches', 'Muffins', 'Kolache',
    'Soft Serve', 'Hash Browns', 'Dunkin’ Bowls'
]

dnkn2 = dnkn1[~dnkn1['Category'].isin(food_categories)].copy()

dnkn2['Category'].value_counts()
```

Out[11]:
```
Category
Hot Coffee            88
Hot Latte             63
Iced Latte            63
Iced Coffee           63
Frozen Coffee         49
Hot Macchiato         45
Hot Cappuccino        43
Iced Cappuccino       42
Iced Macchiato        42
Iced Tea              33
Hot Chocolate         28
Cold Brew Coffee      11
Dunkin’ Refreshers     9
Coolatta               9
Hot Americano          3
Iced Americano         3
Name: count, dtype: int64
```

In [12]:
```
dnkn2 = dnkn2.drop_duplicates()

print("Duplicate rows remaining:", dnkn2.duplicated().sum())
```

```
print("Total rows:", len(dnkn2))
dnkn2
```

```
Duplicate rows remaining: 0
Total rows: 594
```

Out[12]:

	Category	Item	Calories	Total Fat (g)	Saturated Fat (g)	Trans Fat (g)	Cholesterol (mg)	Sodium (mg)	Carbs (g)	Fiber (g)	Sugars (g)	Protein (g)	Weight Watcher Pnts
0	Cold Brew Coffee	Cold Brew Coffee, Large	5	0	0	0	0	20	0	0	0	1	4
1	Cold Brew Coffee	Cold Brew Coffee, Medium	5	0	0	0	0	15	0	0	0	0	5
2	Cold Brew Coffee	Cold Brew Coffee, Small	5	0	0	0	0	10	0	0	0	0	5
3	Cold Brew Coffee	Cold Brew Coffee with Cream, Large	120	12	6	0	35	65	2	0	2	2	126
4	Cold Brew Coffee	Cold Brew Coffee with Cream, Medium	90	9	4.5	0	25	45	1	0	1	2	93.5
...	...	...	...	...	...	...	...	...	...	...	...	...	...
676	Dunkin' Refreshers	Strawberry Dragonfruit Dunkin' Refreshers, Medium	130	0	0	0	0	15	29	0	27	1	156
677	Dunkin' Refreshers	Strawberry Dragonfruit Dunkin' Refreshers, Small	80	0	0	0	0	10	19	0	18	0	98
678	Dunkin' Refreshers	Butter Pecan Swirl Frozen	900	17	13	0	40	570	175	1	162	12	1063

	Category	Item	Calories	Total Fat (g)	Saturated Fat (g)	Trans Fat (g)	Cholesterol (mg)	Sodium (mg)	Carbs (g)	Fiber (g)	Sugars (g)	Protein (g)	Weight Watcher Pnts
		Chocolate, Large											
679	Dunkin' Refreshers	Butter Pecan Swirl Frozen Chocolate, Medium	700	14	10	0	30	430	134	1	124	9	825
680	Dunkin' Refreshers	Butter Pecan Swirl Frozen Chocolate, Small	500	11	8	0	25	290	95	1	87	6	589

594 rows × 13 columns

In [13]:
```
dnkn2 = dnkn2.replace('\xa0', '')
dnkn2 = dnkn2.replace('<5', '')

dnkn2
```

Out[13]:

	Category	Item	Calories	Total Fat (g)	Saturated Fat (g)	Trans Fat (g)	Cholesterol (mg)	Sodium (mg)	Carbs (g)	Fiber (g)	Sugars (g)	Protein (g)	Weight Watcher Pnts
0	Cold Brew Coffee	Cold Brew Coffee, Large	5	0	0	0	0	20	0	0	0	1	4
1	Cold Brew Coffee	Cold Brew Coffee, Medium	5	0	0	0	0	15	0	0	0	0	5
2	Cold Brew Coffee	Cold Brew Coffee, Small	5	0	0	0	0	10	0	0	0	0	5
3	Cold Brew Coffee	Cold Brew Coffee with Cream, Large	120	12	6	0	35	65	2	0	2	2	126
4	Cold Brew Coffee	Cold Brew Coffee with Cream, Medium	90	9	4.5	0	25	45	1	0	1	2	93.5
...	...	...	...	...	...	...	...	...	...	...	...	...	...
676	Dunkin' Refreshers	Strawberry Dragonfruit Dunkin' Refreshers, Medium	130	0	0	0	0	15	29	0	27	1	156
677	Dunkin' Refreshers	Strawberry Dragonfruit Dunkin' Refreshers, Small	80	0	0	0	0	10	19	0	18	0	98
678	Dunkin' Refreshers	Butter Pecan Swirl Frozen	900	17	13	0	40	570	175	1	162	12	1063

	Category	Item	Calories	Total Fat (g)	Saturated Fat (g)	Trans Fat (g)	Cholesterol (mg)	Sodium (mg)	Carbs (g)	Fiber (g)	Sugars (g)	Protein (g)	Weight Watcher Pnts
		Chocolate, Large											
679	Dunkin' Refreshers	Butter Pecan Swirl Frozen Chocolate, Medium	700	14	10	0	30	430	134	1	124	9	825
680	Dunkin' Refreshers	Butter Pecan Swirl Frozen Chocolate, Small	500	11	8	0	25	290	95	1	87	6	589

594 rows × 13 columns

In [14]:
```
print("Non-breaking spaces remaining:", (dnkn2 == '\xa0').sum().sum())
print("<5 values remaining:", (dnkn2 == '<5').sum().sum())
```

```
Non-breaking spaces remaining: 0
<5 values remaining: 0
```

In [15]:
```
import numpy as np

dnkn2 = dnkn2.replace('', np.nan)

dnkn2.isnull().sum()
```

Out[15]:
```
Category                0
Item                    0
Calories                3
Total Fat (g)           3
Saturated Fat (g)       3
Trans Fat (g)           3
Cholesterol (mg)      137
Sodium (mg)             3
Carbs (g)               3
Fiber (g)               3
Sugars (g)              3
Protein (g)             3
Weight Watcher Pnts     3
dtype: int64
```

In [16]:
```
print("Missing values by column:")
print(dnkn2.isnull().sum())

print("Total rows with missing values:", dnkn2.isnull().any(axis=1).sum())
```

```
Missing values by column:
Category                0
Item                    0
Calories                3
Total Fat (g)           3
Saturated Fat (g)       3
Trans Fat (g)           3
Cholesterol (mg)      137
Sodium (mg)             3
Carbs (g)               3
Fiber (g)               3
Sugars (g)              3
Protein (g)             3
Weight Watcher Pnts     3
dtype: int64
Total rows with missing values: 137
```

In [17]:
```
print(dnkn['Cholesterol (mg)'].value_counts().head(15))
```

```
Cholesterol (mg)
0      222
<5     140
20     100
10      53
25      49
30      40
35      33
40      25
15      21
45      11
60       8
55       8
50       7
70       7
100      6
Name: count, dtype: int64
```

In [18]:
```
dnkn3 = dnkn2.dropna().copy()

print("Final dataset shape:", dnkn3.shape)
dnkn3
```

```
Final dataset shape: (457, 13)
```

Out[18]:

	Category	Item	Calories	Total Fat (g)	Saturated Fat (g)	Trans Fat (g)	Cholesterol (mg)	Sodium (mg)	Carbs (g)	Fiber (g)	Sugars (g)	Protein (g)	Weight Watcher Pnts
0	Cold Brew Coffee	Cold Brew Coffee, Large	5	0	0	0	0	20	0	0	0	1	4
1	Cold Brew Coffee	Cold Brew Coffee, Medium	5	0	0	0	0	15	0	0	0	0	5
2	Cold Brew Coffee	Cold Brew Coffee, Small	5	0	0	0	0	10	0	0	0	0	5
3	Cold Brew Coffee	Cold Brew Coffee with Cream, Large	120	12	6	0	35	65	2	0	2	2	126
4	Cold Brew Coffee	Cold Brew Coffee with Cream, Medium	90	9	4.5	0	25	45	1	0	1	2	93.5
...	...	...	...	...	...	...	...	...	...	...	...	...	...
676	Dunkin' Refreshers	Strawberry Dragonfruit Dunkin' Refreshers, Medium	130	0	0	0	0	15	29	0	27	1	156
677	Dunkin' Refreshers	Strawberry Dragonfruit Dunkin' Refreshers, Small	80	0	0	0	0	10	19	0	18	0	98
678	Dunkin' Refreshers	Butter Pecan Swirl Frozen	900	17	13	0	40	570	175	1	162	12	1063

	Category	Item	Calories	Total Fat (g)	Saturated Fat (g)	Trans Fat (g)	Cholesterol (mg)	Sodium (mg)	Carbs (g)	Fiber (g)	Sugars (g)	Protein (g)	Weight Watcher Pnts
		Chocolate, Large											
679	Dunkin' Refreshers	Butter Pecan Swirl Frozen Chocolate, Medium	700	14	10	0	30	430	134	1	124	9	825
680	Dunkin' Refreshers	Butter Pecan Swirl Frozen Chocolate, Small	500	11	8	0	25	290	95	1	87	6	589

457 rows × 13 columns

In [19]:
```
print("Missing values remaining:", dnkn3.isnull().sum().sum())
print("Dataset shape:", dnkn3.shape)
```

```
Missing values remaining: 0
Dataset shape: (457, 13)
```

In [20]:
```
numeric_columns = dnkn3.columns.drop(['Category', 'Item'])

dnkn3[numeric_columns] = dnkn3[numeric_columns].apply(
    pd.to_numeric
)

dnkn3.dtypes
```

Out[20]:
```
Category               object
Item                   object
Calories                int64
Total Fat (g)         float64
Saturated Fat (g)     float64
Trans Fat (g)         float64
Cholesterol (mg)        int64
Sodium (mg)             int64
Carbs (g)               int64
Fiber (g)               int64
Sugars (g)              int64
Protein (g)             int64
Weight Watcher Pnts   float64
dtype: object
```

In [21]:
```
print("Dataset shape:", dnkn3.shape)
print("Missing values:", dnkn3.isnull().sum().sum())
print("Duplicate rows:", dnkn3.duplicated().sum())
print("\nData types:")
print(dnkn3.dtypes)
```

```
Dataset shape: (457, 13)
Missing values: 0
Duplicate rows: 0

Data types:
Category               object
Item                   object
Calories                int64
Total Fat (g)         float64
Saturated Fat (g)     float64
Trans Fat (g)         float64
Cholesterol (mg)        int64
Sodium (mg)             int64
Carbs (g)               int64
Fiber (g)               int64
Sugars (g)              int64
Protein (g)             int64
Weight Watcher Pnts   float64
dtype: object
```

In [22]:
```
dnkn3.to_csv("BUS303_Dunkin_Cleaned.csv", index=False)

print("Cleaned dataset saved successfully!")
```

```
Cleaned dataset saved successfully!
```

In [23]:
```
dnkn4 = pd.read_csv("BUS303_Dunkin_Cleaned.csv")

print("Reloaded dataset shape:", dnkn4.shape)
print("Missing values:", dnkn4.isnull().sum().sum())
print("Duplicate rows:", dnkn4.duplicated().sum())
```

```
Reloaded dataset shape: (457, 13)
Missing values: 0
Duplicate rows: 0
```

For Deliverable Two, I worked with the Dunkin' Donuts nutritional dataset to identify and clean issues that could affect data analysis. I started by reviewing the dataset's structure, checking for missing values and duplicate records, and examining the different food and beverage categories. I also reviewed the nutritional columns to identify values that needed cleaning.

During the cleaning process, I removed selected food categories, eliminated duplicate records, and handled missing and nonnumeric values. I also converted the nutritional columns into numeric data types so they could be used for future analysis.

After completing the cleaning process, my final dataset contained 457 rows and 13 columns, with no missing values or duplicate records. I then saved the cleaned dataset as a CSV file and reloaded it to verify that the file was saved correctly. The cleaned dataset is now ready for further analysis.